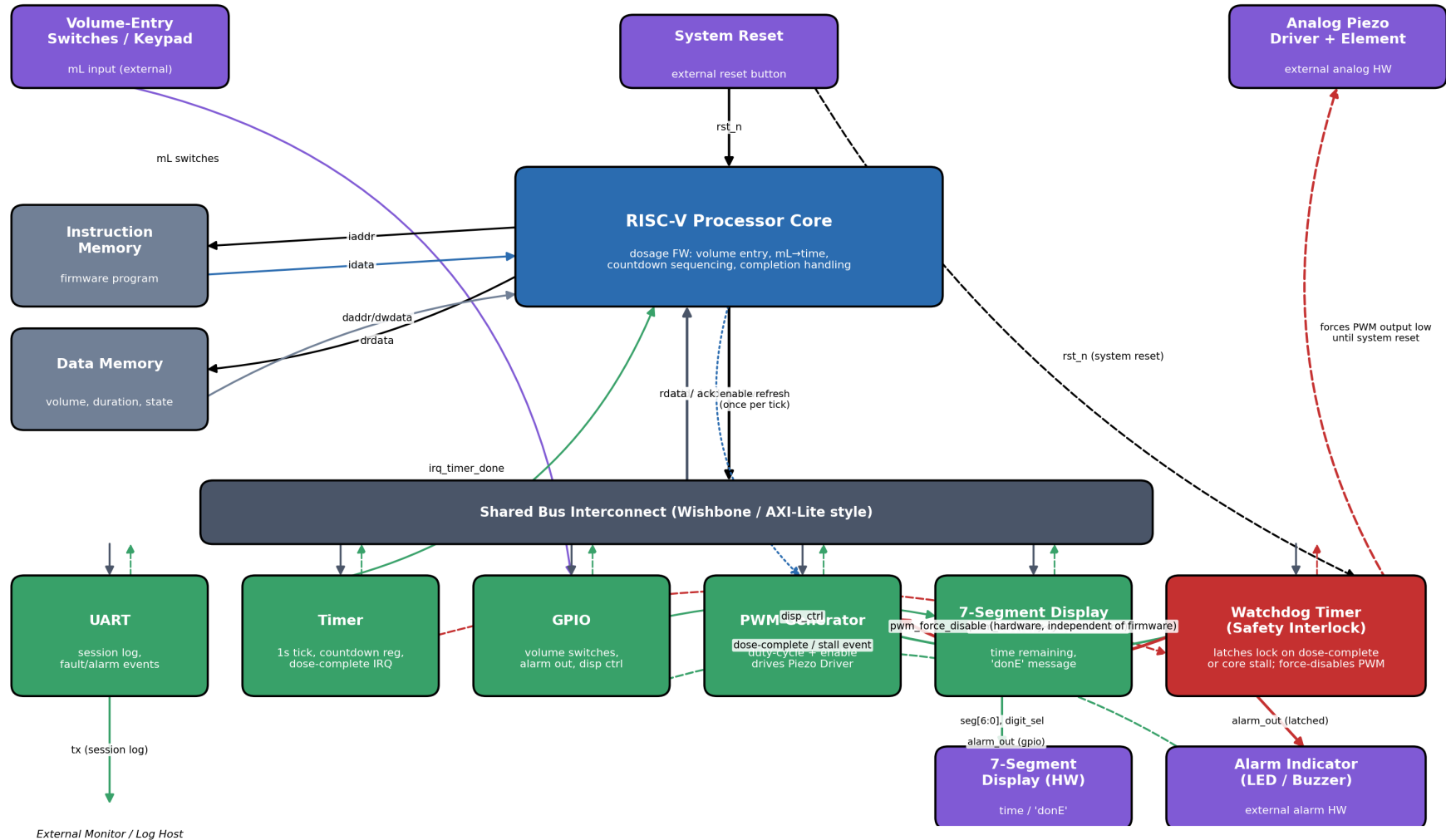


NebuCore SoC

RISC-V-Based System-on-Chip for Volume-to-Time Dosage Control and Safety-Interlocked Nebulizer Actuation — Block Diagram & Signal List

NebuCore SoC — Top-Level Block Diagram

RISC-V-Based SoC for Volume-to-Time Dosage Control and Safety-Interlocked Nebulizer Actuation



Signal List

Signals are grouped by block. Direction is given from the perspective of the named block. "Bus" indicates the signal is carried as part of the shared memory-mapped bus interconnect (address, write-data, read-data, write-enable, chip-select, acknowledge) between the RISC-V core and every peripheral; only bus signals with block-specific meaning are broken out separately below.

1. RISC-V Processor Core (top-level control ports)			
Signal	Width	Dir.	Description
clk	1	in	System clock.
rst_n	1	in	Active-low system reset, sourced from the external Reset button; also broadcast to the Watchdog Timer so a manual reset can clear a latched lock.
bus_addr	32	out	Address driven onto the shared bus for the current memory or peripheral access.
bus_wdata	32	out	Write-data driven onto the shared bus (register writes to peripherals, data-memory stores).
bus_rdata	32	in	Read-data returned from the addressed memory or peripheral.
bus_we	1	out	Bus write-enable; qualifies bus_wdata as a valid write.
bus_sel	1	out	Bus byte/word select strobe for the current transaction.
bus_ack	1	in	Bus acknowledge / ready from the addressed slave, used to extend transactions where needed.
irq_timer_done	1	in	Interrupt from the Timer peripheral: fires each 1-second tick and again at dose-complete (count = 0).
irq_wdt	1	in	Optional status interrupt from the Watchdog Timer (e.g., core-stall detection), used for logging; the lock itself is enforced in hardware independent of this line.

2. Instruction Memory			
Signal	Width	Dir.	Description
iaddr	32	in	Instruction fetch address driven by the RISC-V core.
idata	32	out	Instruction word returned to the core's fetch stage.
ien	1	in	Fetch enable / chip-select from the core.

3. Data Memory			
Signal	Width	Dir.	Description
daddr	32	in	Data address for load/store of entered volume, computed duration, and session state.
dwwdata	32	in	Write data for stores from the core.
drdata	32	out	Read data returned to the core for loads.

dwe	1	in	Data memory write-enable.
dre	1	in	Data memory read-enable.

4. Shared Bus Interconnect

Signal	Width	Dir.	Description
m_addr / m_wdata / m_we	32/32/1	in	Master-side (core) address, write-data, and write-enable presented to the interconnect.
m_rdata / m_ack	32/1	out	Read-data and acknowledge returned to the core.
s_sel[N-1:0]	N	out	Per-peripheral chip-select/decode lines (one per attached IP: UART, Timer, GPIO, PWM Generator, 7-Segment Display Controller, Watchdog Timer).
s_addr / s_wdata / s_we	32/32/1	out	Address, write-data, and write-enable forwarded to the selected peripheral's register interface.
s_rdata / s_ack	32/1	in	Read-data and acknowledge muxed back from whichever peripheral is currently selected.

5. UART

Signal	Width	Dir.	Description
tx	1	out	Serial transmit line to the external monitor/log host; carries session start, completion, and fault/alarm log messages.
rx	1	in	Serial receive line (reserved for future host commands / diagnostics).
tx_data[7:0]	8	in	Byte to transmit, written by firmware via the bus.
tx_valid / tx_ready	1/1	in/out	Handshake: firmware asserts tx_valid to load a byte; peripheral asserts tx_ready when the transmit buffer is free.
rx_data[7:0] / rx_valid	8/1	out	Received byte and its valid strobe, readable by firmware via the bus.

6. Timer

Signal	Width	Dir.	Description
timer_load[15:0]	16	in	Countdown value (in seconds) loaded by firmware after the mL-to-time conversion.
timer_en	1	in	Countdown enable, set by firmware to start/stop the session timer.
timer_count[15:0]	16	out	Live countdown value, read by firmware once per tick to refresh the display.
tick_1s	1	internal	Internal 1 Hz strobe that decrements timer_count (not bus-visible; listed for clarity of timing behavior).
irq_done	1	out	Asserted when timer_count reaches zero — the dose-complete event that triggers PWM disable and the Watchdog lock.

7. GPIO

Signal	Width	Dir.	Description
gpio_in[7:0]	8	in	Volume-entry switch / keypad inputs (mL value), sampled by firmware.
gpio_out[7:0]	8	out	General-purpose outputs: display control strobes and the (pre-latch) alarm indicator line.
gpio_dir[7:0]	8	in	Per-pin direction configuration register.
gpio_irq	1	out	Optional interrupt-on-change for the volume-entry inputs, used instead of polling.

8. PWM Generator

Signal	Width	Dir.	Description
duty_cycle[7:0]	8	in	Duty-cycle register, set by firmware to control nebulizing intensity.
pwm_en	1	in	Enable bit; firmware must keep re-asserting this once per Timer tick, so a core stall lets the enable lapse rather than leaving the actuator running.
pwm_out	1	out	PWM drive waveform sent to the external analog Piezo Driver.
pwm_force_disable	1	in	Hardware override input driven directly by the Watchdog Timer; when asserted, forces pwm_out low regardless of pwm_en or firmware state.

9. Seven-Segment Display Controller

Signal	Width	Dir.	Description
digit_data[15:0]	16	in	Value to display (countdown seconds, or the encoded 'donE' completion message), written by firmware.
disp_en	1	in	Display-update enable/strobe, pulsed by firmware once per Timer tick.
seg_out[6:0]	7	out	Segment drive lines (a–g) to the physical seven-segment display.
digit_sel[3:0]	4	out	Digit-select / multiplex lines for a multi-digit display.

10. Watchdog Timer (Safety Interlock)

Signal	Width	Dir.	Description
wdt_trigger	1	in	Asserted by firmware (or derived directly from Timer irq_done) on dose completion, arming the lock.
wdt_stall_detect	1	internal	Independent stall/timeout detection: if the core fails to service the session within its expected window, the Watchdog latches the same locked state without firmware involvement.
wdt_locked	1	out	Latched lock-status flag; once set, remains set regardless of any subsequent bus or firmware activity.
pwm_force_disable	1	out	Hardware line into the PWM Generator that forces pwm_out low while wdt_locked is set (see PWM Generator, above).
alarm_out	1	out	Drives the alarm indicator (LED/buzzer) directly, overriding the GPIO-driven alarm path, for as long as the lock is latched.

wdt_reset_clear	1	in	Tied to system rst_n; this is the only way to clear wdt_locked and re-enable the PWM Generator.
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11. External / Off-Chip Signals

Signal	Width	Dir.	Description
Volume-entry switches / keypad	8	→ GPIO	Caregiver-entered liquid medication volume, in mL.
System reset button	1	→ core, → WDT	Manual reset; the only way to clear a latched Watchdog lock and start a new session.
Piezo Driver input	1	← PWM	Analog driver stage for the piezoelectric nebulizing element; receives pwm_out and is force-held low by the Watchdog after dose completion.
Seven-segment display (physical)	7 + digit sel	← 7-seg ctrl	Shows time-remaining during countdown and the 'donE' message at completion.
Alarm indicator (LED/buzzer)	1	← GPIO / ← WDT	Signals session completion / lockout state to the caregiver.